

ASSIGNMENT – OBJECT ORIENTED PROGRAMMING WITH JAVA (BCS-403)

Time: Self-paced
Maximum Marks: 100

Mandatory Submission Guidelines

Assignment must be:

- **Handwritten**
- Properly scanned and uploaded as **PDF only**
- File naming format (STRICT): RollNo_Name_OOPS_Assignment.pdf

SECTION A – THEORY *(Attempt all questions)*

Q1. (5 Marks) Explain JVM, JRE, and JDK with suitable diagram.

Q2. (5 Marks) Explain the concepts of **Encapsulation** and **Polymorphism** with examples.

Q3. (5 Marks) Explain the **Exception Handling mechanism** in Java. Also differentiate between **checked and unchecked exceptions**.

Q4. (5 Marks) Explain **Java Collection Framework**. Differentiate between **List, Set, and Map**.

SECTION B – PROGRAMMING *(Attempt all questions. Show output for each program.)*

Q5. (10 Marks)

Write a Java program to demonstrate Class, Object, Constructor, and Method.

Q6. (10 Marks)

Write a Java program to demonstrate Inheritance and Method Overriding.

Q7. (10 Marks)

Write a Java program to handle exceptions using try-catch-finally and demonstrate division by zero.

Q8. (10 Marks)

Write a Java program to create an ArrayList, add elements, remove elements, and display size and contents.

Q9. (10 Marks)

Write a Java program using both Lambda Expression and Functional Interface to perform an addition of two numbers.